Cupel: A Framework for Professional Competence Provenance

Verifying Human Expertise in a World of Capable Machines
Position Paper and Framework Specification — v0.4

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Abstract

The deployment of autonomous and semi-autonomous artificial intelligence agents into professional workflows has created a structural verification problem in the human layer those systems depend on. Two trends are converging. First, AI systems can now produce the signals of professional competence — passing licensing examinations, generating credible portfolios, drafting credentialed work — without possessing the judgment, accountability, or contextual experience that credentials were designed to indicate. Second, the apprenticeship pipeline through which qualified human supervisors are produced is contracting: entry-level technology hiring at the top fifteen firms fell 25% between 2023 and 2024 [2], and UK graduate technology roles fell 46% in 2024 [3]. These trends compound: organisations are deploying systems faster than they can identify the humans capable of supervising them, while simultaneously narrowing the pipeline that produces such humans. This paper identifies the resulting gap, characterises the property of competence that matters in agent-mediated work — what we term competence *below the abstraction layer* — and proposes Cupel, an open framework specification for professional competence provenance. Cupel defines a taxonomy of five trust signal types, a JSON-LD schema for representing and linking those signals, and an interoperability architecture that sits above existing standards including W3C Verifiable Credentials, C2PA, the Credential Transparency Description Language (CTDL), and Open Badges 3.0. This revision adds an explicit treatment of the human stakes behind the pipeline data: early 2026 labour-market evidence shows the entry ramp into knowledge work narrowing before any broad unemployment signal appears, which changes what an adequate policy and infrastructure response looks like. The framework is released under Apache 2.0, with trademark protection ensuring conformance integrity.

Contents

1 Introduction

When an autonomous agent files a tax return, drafts a contract, recommends a clinical course of action, or commits production code, the integrity of that output depends on a qualified human who can detect its errors, intervene meaningfully, and accept professional responsibility. The credential system that organised professional labour markets for most of the twentieth century was designed in part to identify those humans: a chartered accountant, licensed physician, or admitted barrister was not merely a knowledge marker but an accountability node embedded in a system of professional bodies, regulatory oversight, and liability mechanisms.

This identification system is failing on two fronts simultaneously, and the failure is accelerating.

The signal is no longer informative. The effectiveness of credentials as competence proxies depended on a single equilibrium condition: *gaming the proxy must be roughly as hard as becoming genuinely competent*. This condition has been violated. In September 2025, large language models passed CFA Level III mock examinations including written essays, with leading models scoring above 79% [1]. The same pattern has emerged across the bar exam, medical licensing examinations, and engineering certification assessments. AI systems can acquire the signal without acquiring the substance. This is a textbook instantiation of Goodhart's Law: when a measure becomes a target, it ceases to be a good measure [7]. The market response — issuing more credentials, from 334,114 unique credentials in the United States in 2018 to 1.85 million by 2025 [8] — has increased signal volume while continuing to degrade signal quality. Gartner projects that one in four candidate profiles worldwide will be entirely fabricated by 2028, and 41% of enterprises report having already onboarded a fraudulent candidate [9].

The pipeline that produces qualified supervisors is contracting. Entry-level technology hiring at the top fifteen firms fell 25% between 2023 and 2024 [2]; UK graduate technology roles fell 46% in 2024 [3]. Junior roles are not interchangeable with senior ones — they are the structured apprenticeship layer through which senior practitioners are formed. Organisations compressing junior hiring are defunding their own future supervisory capacity. The problem is time-delayed: by the time the shortage of qualified supervisors becomes acute, the cohort that would have filled those roles will not exist.

Deployment is outpacing oversight. Multi-agent system adoption rose 327% in four months [5]. Only one in five companies has mature governance for autonomous agents [4]. 78% of AI users bring personal tools to work, often outside any oversight framework [6]. The European Union's AI Act activates high-risk compliance obligations through 2026, with Article 14 requiring meaningful human oversight and Article 50 imposing content provenance obligations [18]. These obligations presuppose a verifiable population of competent supervisors. No mechanism currently exists to identify that population at scale.

The problem is structural, not cyclical. The signal layer of professional labour markets and the supervisory layer for AI deployment have become the same problem: organisations need to know which humans can genuinely catch the errors that machines now produce at scale. This paper describes the infrastructure that closes that gap: the Cupel Professional Competence Provenance Framework.

2 Below the Abstraction Layer

A central claim of this paper is that the property of competence which now matters in agent-mediated work is not well captured by existing credentialing frameworks. We call this property competence *below the abstraction layer*.

Most professional credentials measure performance *above* the AI abstraction layer — on tasks that AI systems can now also perform. A candidate who can pass the CFA Level III demonstrates an ability that a large language model also demonstrates. The credential, on its own, no longer discriminates between the two. Performance at this level has become diagnostic of nothing more specific than the ability to produce credentialed output.

Competence below the abstraction layer is qualitatively different. It is the ability to:

- recognise when an AI output is wrong, in cases where the error is non-obvious to a non-expert evaluator;
- identify the specific dimension on which the output is wrong (factually, structurally, contextually, ethically);
- determine what corrective action is appropriate and, where applicable, execute it;
- accept professional responsibility for the corrected output under the relevant accountability framework.

Three illustrative cases clarify the distinction:

A clinician using an AI system to read a diagnostic scan is not abdicating responsibility to the system. The clinician retains diagnostic authority. The signal that matters is whether the clinician can recognise a subtle finding that the model missed or misclassified — not whether the clinician could replicate the model's general performance on a licensing examination.

A code reviewer approving an AI-generated patch must be able to identify a security vulnerability that the model did not flag. The reviewer's competence is judged by what they catch, not by what they would produce unassisted.

A compliance officer signing off on an AI-drafted disclosure must understand the regulatory rule the AI applied, the conditions under which that rule has exceptions, and whether the present case falls within those exceptions. The signature is meaningful only if the officer has the capacity to dissent.

In each case, the human is exercising what regulators term "human oversight" or "human in the loop" [18]. The legal and ethical force of that oversight depends entirely on whether the human can in fact catch the loop's mistakes — a property that traditional credentials were not designed to verify, and that AI-era credential gaming makes harder to verify by surface inspection.

Cupel's design centres on producing legible signals of competence below the abstraction layer. The five trust signal types defined in Section 5 are each ways of evidencing this property; the multi-signal model is the framework's principal answer to the question of how such competence can be verified at scale without recreating the gameable conditions of single-signal credentialing.

3 The Human Stakes

The pipeline data cited in Section 1 is not merely a supply-side constraint on future supervisory capacity; it describes a present and specific labour-market anxiety, and the distinction matters for what an adequate response looks like.

Anthropic's Economic Index reports that, as of early 2026, hiring of 22–25 year-olds into the occupations most exposed to AI has slowed by approximately 14%, while no clear broad unemployment signal has yet appeared in those same occupations [19]. Roughly one in five workers in AI-exposed jobs report concern about economic displacement, with software engineers among the most concerned of any profession surveyed [19]. Read together, these findings indicate that the entry ramp into knowledge work is narrowing *before* any headline unemployment figure moves, because employers can defer hiring decisions long before they take the more visible step of reducing headcount.

This distinction is consequential. If the operative crisis were simply contracting employment, the appropriate policy response would be conventional: income support and retraining. But if the operative crisis is that the apprenticeship structure converting novices into accountable experts is quietly failing — junior roles being the mechanism by which practitioners historically acquired the below-the-abstraction-layer competence characterised in Section 2 — then income support and retraining do not address the underlying failure. What is additionally required is verification infrastructure that can recognise competence in practitioners who did not pass through that structure in the traditional way, and that can make legible who, among a shrinking cohort, is actually capable of supervising agent-mediated work.

This also reframes who has standing in the conversation this paper is part of. The verification problem is not confined to standards bodies, credentialing vendors, and enterprise HR functions. It is a live concern for anyone whose career ladder has had rungs removed, and for anyone deciding how much unsupervised authority to extend to a system that no one has yet verified a human can adequately oversee. A parallel and growing concern, documented separately, is that AI has made confident commentary about these dynamics cheap to produce and correspondingly easy to overtrust, widening the gap between people describing the future of knowledge work and people building a working response to it [20]. This paper, and the open framework it proposes, is offered as evidence-based participation in that response rather than commentary on it.

4 What Credentials Actually Did

Understanding what must replace credentials requires clarity on what they actually provided. Four distinct functions are now unbundling:

Information asymmetry resolution. Credentials were trust shortcuts for principals who cannot evaluate agent competence directly. If AI can produce competent-seeming professional output, the principal now faces *worse* information asymmetry: they cannot determine whether the professional is applying judgment on top of AI output or simply relaying it.

Accountability infrastructure. Professional credentials were embedded in systems of licence revocation, malpractice liability, and disciplinary action. AI-generated output carries no equivalent accountability infrastructure. The credential was the entry condition for accountability, not merely a knowledge signal.

Community of practice gatekeeping. Much professional expertise is tacit knowledge distributed within practice communities. The credential was the admission ticket. This gatekeeping function persists even as the knowledge-signalling function collapses, creating an increasingly hollow dynamic.

Scalable evaluation. Credential verification scales: a recruiter can screen 10,000 applicants by checking degrees. Any replacement system must address scalability. Most proposed alternatives — portfolio review, work samples, structured reference checks — are high-bandwidth but low-scale. This asymmetry is the central engineering challenge.

5 Emerging Trust Mechanisms

Five new trust mechanisms are forming at different stages of maturity. The Cupel framework is designed to provide infrastructure for all five.

5.1 Demonstrated Outcome Trails

Verifiable records of actions taken and outcomes achieved: portfolios, open-source contributions, competition results, documented client outcomes, and public track records. This is the most mature emerging mechanism. Its limitation is domain specificity — outcome trails that are legible in software engineering (GitHub) or data science (Kaggle) have no equivalent in consulting, legal practice, or financial advisory.

5.2 Behavioural Consistency Signals

Consistency across time and contexts as a primary trust signal. A professional with a decade of context-appropriate judgment documented across engagements signals something a certification cannot. Consistency is hard to fake because it requires sustained performance. It is also hard to *capture* without infrastructure purpose-built for that function.

5.3 Network-Verified Reputation

Professional networks becoming the primary trust mechanism rather than a supplement. 63% of consumers identify verified reviews as the most important trust signal [11]. In professional contexts this manifests as referral networks, public endorsements, and the "who vouches for you" graph. The limitation is susceptibility to network homophily and collusion.

5.4 Process Transparency and Human-AI Collaboration Records

The new credential is not "I know finance" but "here is how I use AI in my process, here is where I override it, and here is my track record of those overrides being correct." Trust shifts from *what you know* to *how you think* alongside AI systems. This is nascent but structurally important as AI capability continues to advance.

A critical property of this signal type is that it captures *attribution clarity* — not a ratio of human versus machine effort, but a record of at which decision points human judgment was exercised, what alternatives were available, and what the consequences were. A practitioner who delegates execution to AI while making all structural choices is exercising full professional accountability; the relevant signal is the quality and track record of those structural choices, not the volume of manual work.

5.5 Provenance and Attribution Infrastructure

Content credentials, human-in-the-loop documentation, and chain-of-custody metadata for professional work product. The term *content credentials* follows C2PA's established nomenclature for cryptographically-bound provenance metadata attached to a digital artefact [12]. The analogy is organic food labelling: a mechanism for distinguishing human-exercised judgment from AI-generated output at the level of individual work products. This layer is being accelerated by regulatory requirements including EU AI Act Article 50 content provenance obligations and the California AI Transparency Act (disclosure requirements effective January 2026).

Critically, provenance-tracked signals must capture an *AI audit trail* — a structured log of process metadata — in addition to final outputs. The term "AI audit trail" is now established in both regulatory compliance contexts and enterprise AI governance frameworks [15]. An AI system can produce a polished final deliverable that is indistinguishable from expert human output. What AI cannot replicate authentically is the iterative trail of a human-led process: the drafts, corrections, mid-course reversals, and contextually-situated judgments that characterise genuine professional engagement with a problem. A provenance-tracked signal for a piece of professional work should therefore include, at minimum: the content credential (which AI systems, if any, were used and in what capacity, expressed in a C2PA-compatible format), the AI audit trail (evidence of human decision points across the development of the work), and the accountability anchor (who is responsible for the outcome and under what professional framework).

5.6 The Structural Tension

Old credentials were *scalable but low-bandwidth.* New trust mechanisms are *high-bandwidth but low-scale.* This asymmetry creates market bifurcation: high-value professional work will operate on rich, verified trust signals; routine work will be commoditised into AI pipelines with binary quality checks. The middle layer of professional practice faces the most disruption.

6 Market Landscape and the Gap

The current market for professional trust infrastructure can be characterised as five layers, each with distinct players and commercial dynamics.

Layer	Representative Players	Maturity
Identity Verification & Fraud Detection	Yardstik, GetReal Security, Pindrop, Sherlock AI	Most funded; pain-driven buying
Skills Assessment & Validation	TestGorilla, HireVue, Codility, Pymetrics	Growing fast; 41% of firms now use assessments
Digital Credential Infrastructure	Credential Engine (CTDL), MIT DCC, 1EdTech (CLR)	Standards-building; technically strong, commercially early
Content Provenance & Authenticity	C2PA coalition (200+ members inc. Adobe, Microsoft, Google)	Regulatory-driven; ISO fast-track underway
Professional Reputation & Outcome Tracking	LinkedIn (self-reported), Crosschq, Toptal, Kaggle	Least developed; biggest structural gap

Table 1: The five-layer professional trust infrastructure landscape.

Most investment is concentrated in Layers 1 and 2: detecting fraud and testing skills. These are defensive plays. The *constructive* infrastructure — systems that positively identify competence and track demonstrated judgment over time — is severely underdeveloped.

The critical absence is a **cross-cutting evaluation and interoperability layer**: a common language for describing, linking, and evaluating professional competence signals across the entire fragmented ecosystem. With 1.85 million credentials in circulation and widespread inability to evaluate them, this aggregation layer represents the most significant unmet structural need in professional labour markets.

The logical convergence point is a unified professional competence provenance system that connects all five layers into a coherent, interoperable whole. Whoever defines the architecture for this convergence is building the replacement for the degree.

7 The Cupel Framework

7.1 Design Rationale: Open Standard Over Product

The Coalition for Content Provenance and Authenticity (C2PA) provides the most instructive precedent. C2PA succeeded not by launching a product but by publishing a specification and building a coalition. It is now fast-tracked as an ISO standard and has over 200 member organisations. The professional competence provenance problem is structurally similar: it requires cross-industry adoption, trust from multiple competing stakeholders, and interoperability between platforms with conflicting commercial interests. An open standard maximises adoption and positions the framework as neutral infrastructure rather than as a competitor to existing players.

7.2 Framework Components

Cupel specifies five interrelated components:

Trust Signal Taxonomy. A structured classification of professional competence signals into five types:

- *Credential-based*: issued qualifications, licences, certifications
- *Assessment-based*: structured evaluations of demonstrated skill
- *Outcome-based*: verifiable records of actions and consequences
- *Peer-verified*: endorsements and attestations from identified third parties
- *Provenance-tracked*: content credentials and AI audit trails for professional work product, capturing both chain-of-custody metadata and the process record (AI tool disclosure, human decision points, iteration history)

The five-type model is designed for signal *diversity* — what practitioners in skills-based hiring increasingly term a *trust portfolio* [17]: no single signal type is sufficient, and the evidential weight of a competence claim increases with corroboration across multiple independent signal types. This design makes the framework resistant to single-vector gaming: inflating peer-verified signals, for example, does not compensate for absence of outcome-based or provenance-tracked evidence.

Core Schema. A JSON-LD schema for expressing trust signals and their relationships, aligned with the W3C Verifiable Credentials Data Model v2.0 [13]. The schema captures signal type, issuer identity, subject identity, temporal scope, domain specificity, and verification method. For provenance-tracked signals specifically, the schema includes a `contentCredential` field (C2PA-compatible declaration of AI tools used and their scope of involvement) and an `auditTrail` field (structured log of human decision points and process iterations). These fields are optional for other signal types but required for provenance-tracked signals.

Selective Disclosure Principle. The framework defines a vocabulary for describing competence signals; it does not mandate a surveillance architecture. Conformant implementations must support selective disclosure: a practitioner controls which signals are shared, with which parties, and for which purposes. This principle is consistent with W3C Verifiable Credentials' selective disclosure capabilities and with the EU General Data Protection Regulation's data minimisation requirements. Provenance tracking creates a potential privacy tension — detailed process logs could expose proprietary methods or working patterns — and the selective disclosure principle is the framework's structural answer to that tension.

Signal Quality Criteria. Evaluation dimensions for assessing the evidential weight of individual signals: source credibility, recency, verifiability, specificity, and consistency with concurrent signals.

Interoperability Architecture. Defined mappings between Cupel signal types and existing standards, enabling bidirectional translation without requiring issuing platforms to adopt the full framework.

Governance Model. A trust list maintenance mechanism, conformance programme, and contributor framework modelled on established open standards governance (W3C, IETF, C2PA).

The governance model must specifically address what may be termed the *oracle problem of attribution*: if Cupel relies on issuing systems (certification bodies, HR platforms, assessment providers) to supply signal metadata, the framework's trust guarantees are bounded by the integrity of those systems. A dishonest or negligent issuer can supply structurally conformant but substantively false signals.

The framework addresses this through an *issuer trust registry*: a reputation mechanism in which issuers accumulate credibility based on the observable real-world correlation between their signals and verified outcomes over time. The term "trust registry" is established in W3C Verifiable Credentials contexts and in MIT's Digital Credentials Consortium Governance Framework for Issuer Identity Registries (June 2025) [16]. This is a market mechanism, not a technical mandate. It follows the pattern of email sender reputation systems (SPF/DKIM scoring) and OpenID Federation's trust chains: no central authority assigns credibility; credibility emerges from the longitudinal track record of each issuer's signals. Evaluators weight signals from high-integrity issuers more heavily, and the market incentivises issuers to maintain signal quality. The registry itself requires ongoing data collection and curation infrastructure; the specification defines the interface but does not mandate a single operator.

7.3 Relationship to Existing Standards

Cupel does not replace existing standards. It sits above them as an evaluation and interoperability layer:

Standard	Role in Cupel Architecture
W3C Verifiable Credentials	Cryptographic verification layer for all signal types
C2PA	Content provenance architecture; template for work product provenance signals
Credential Engine / CTDL	Structured description of credential-based signals
Open Badges 3.0 / CLR	Micro-credential and learner record formats for assessment-based signals
ISO/IEC 27001, 29101	Security and privacy frameworks for implementation conformance

Table 2: Cupel's relationship to existing standards.

The novel contribution is the cross-cutting layer: a unified representation that allows a composite professional competence provenance profile to be constructed from signals originating in any of the above ecosystems.

8 Licensing and Conformance

The framework specification and core schema are released under Apache 2.0. Any implementation may use the technical content without restriction. The Cupel name and logo are trademarked (UK00004352899, filed 11 March 2026), ensuring that conformance claims carry meaning: only implementations that meet published conformance criteria may identify themselves as Cupel implementations. This follows the Linux and OpenID precedent, where an open technical artefact is paired with a protected name to prevent the dilution of conformance through unverified implementations.

The selection of Apache 2.0 in preference to a copyleft licence reflects an adoption-first strategy. The framework's value depends on broad cross-industry uptake; a permissive licence removes friction for commercial implementers and standards bodies that would be unable to adopt a strong copyleft licence as a matter of internal policy. The specification is intended to be usable without restriction by both open and proprietary implementations.

Specific operational infrastructure that supports the framework — for example, an issuer trust registry that aggregates real-world signal-outcome correlations — may be operated by independent parties under their own governance arrangements. The specification defines what such infrastructure must do to be conformant; it does not nominate a single operator and does not preclude any particular funding model.

9 Conclusion

The credential system that organised professional labour markets for most of the twentieth century is structurally compromised, and the supervisory layer that AI deployment depends upon rests on the same compromised foundation. The cause is not a failure of operators but the convergence of three trends: AI systems that can produce the surface signals of competence without the substance, a contracting apprenticeship pipeline that is narrowing the cohort of practitioners from which future supervisors will be drawn, and an accelerating deployment of agents into roles that presuppose competent human oversight.

The market response to date has concentrated on defensive infrastructure: detecting fraud, testing skills in controlled environments, verifying that AI outputs meet quality thresholds. The constructive work of building new positive trust infrastructure — systems that identify and track the humans capable of supervising agent work over time — is largely undone.

Cupel proposes the architecture for that infrastructure. It is released as an open standard because the problem requires coalition, not competition. The framework's contribution is deliberately bounded: it specifies a common vocabulary, a data format, and an interoperability architecture. It does not assert what makes someone competent; it provides the substrate on which competence claims from many sources can be expressed, linked, and evaluated together. The specification, reference implementation in development, and governance framework are available at github.com/vdmeu/CUPEL and cupel.foundation.

The framework's adoption strategy targets the alternative credential and micro-assessment ecosystem as the initial cohort: coding bootcamps, AI-tutor platforms, portfolio-based assessment providers, and open-source contribution tracking tools. These organisations have the most to gain from interoperability — their signals are currently unreadable to enterprise hiring systems — and no incumbent lock-in to protect. Broad adoption in the long tail of the credential ecosystem creates the interoperability pressure that eventually compels larger platforms to participate, mirroring the bottom-up adoption dynamics of C2PA and Open Badges. Regulatory tailwinds — particularly the EU AI Act's transparency and oversight requirements — provide additional leverage with larger incumbents for whom compliance creates a structural incentive to adopt standardised provenance formats.

Several open questions remain, and the framework is offered for community engagement rather than as a finished proposal. Among them: the technical specification of a live provenance signal data model; the operational design of selective disclosure under the General Data Protection Regulation and equivalent regimes; the resistance properties of the multi-signal model against sophisticated gaming strategies; and the appropriate governance arrangements for an issuer trust registry. Community input on the taxonomy, schema, and interoperability mappings is welcomed through the repository's issue tracker and discussion board.

About the Author

Eugene Andrie Merwe-Chartier writes and maintains Cupel as an open framework and an ongoing, evidence-based public argument, not as a commercial product. Correspondence, critique, and contributions are welcome at contact@cupel.foundation and github.com/vdmeu/CUPEL, where periodic commentary tracking the framework's three constituent threads — provenance, credentialing, and the human impact of automated knowledge work — is also published.

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the intellectual origin of this work.

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